

Extended Abstract

Motivation Safe RL methods often train policies with explicit cost constraints, but the safety of these methods during deployment can still fail as a result of distribution shift. The aim of this project is to ask whether a separate risk model can predict when a trained safe RL policy is about to violate a safety budget and whether the resulting probability can gate the policy with a conservative fallback (if risk is too high, fall back to a less risky alternative). This is studied in Safety-Gymnasium’s `SafetyPointGoal1-v0` where a point robot must reach goals while avoiding hazards. The objective is not to produce a state-of-the-art safe RL agent but to build a clean end to end falsification and risk-scoring pipeline with interpretable baselines.

Method The base policy is trained as a constrained Markov decision process using PPO-Lagrangian. The policy maximizes reward subject to a cost limit per episode of 25 and learns both a reward critic $V(s)$ and a cost critic $V_c(s)$. Since $V_c(s)$ already makes an estimate of upcoming cost it is the main baseline risk score. It is compared to a learned risk head that receives only runtime available features (these are observation, policy action mean, policy log standard deviation, $V(s)$, $V_c(s)$ and recent cost over the last ten steps). The short horizon label at time t is $y_t = 1$ when the cost accumulated over the next $H = 50$ steps exceeds the chosen risk budget B .

Implementation Distribution shifts are implemented as Gymnasium wrappers. These are Gaussian observation noise for sensor disturbance, Gaussian action noise for agent disturbance and action scaling for an environment or dynamics disturbance. Rollouts under these disturbance distributions form a direct sampling falsification dataset. Each episode is a sampled trajectory and each positive window is a failure of the short horizon safety specification. The learned risk head is a small two layer MLP while the cost critic baseline is calibrated with 1D Platt scaling so both methods output probabilities.

Results The effective automatically selected risk budget is $B = 0.5$. On the in-distribution shifted test split, the learned risk head improves AUROC from 0.907 for Platt-scaled $V_c(s)$ to 0.956 and improves AUPRC from 0.466 to 0.751. Family level results show the largest AUROC improvement on action noise (+0.394), observation noise (+0.255) and nominal rollouts (+0.527). On the held-out action-scale disturbance, the risk head gets AUROC 0.530 vs 0.387 for the baseline. These results validate the pipeline and show that the learned head can extract useful signal beyond the cost critic.

Discussion and Success Criterion The minimum result that counts as success is a riskhead improvement of at least 0.10 AUROC over Plattscaled $V_c(s)$ on the heldout disturbance family. The action scale result meets this criterion with a 0.143 AUROC gain, which is meaningful because this disturbance family is not used to train the risk head. At the same time absolute OOD AUROC is only 0.530, so the result is better described as useful but limited generalization rather than a solved OOD safety monitor. The main takeaway is that a learned deployment-time head can add signal beyond the safe RL cost critic. However selective deployment also depends heavily on calibration and on the fallback action being reasonable.

Deployment-Time Risk Scoring for a Safe RL Agent under Distribution Shift

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Abstract

This project investigates whether a separate risk model can predict, at deployment time, when a trained safe reinforcement learning policy is likely to violate a safety budget under distribution shift. The base policy is trained with PPO-Lagrangian on Safety-Gymnasium’s `SafetyPointGoal1-v0`. It gives both a reward critic and a cost critic. The cost critic $V_c(s)$ is treated as the primary baseline risk score and compared to a learned risk head that combines $V_c(s)$ with policy distribution features, reward value, observation and recent cost. Distribution shifts are modeled as disturbance distributions. These are observation noise, action noise and action scaling. The resulting shifted rollouts are direct sampling falsification data, where the labels come from short horizon cost window violations. The risk head improves in-distribution AUROC from 0.907 to 0.956 over the calibrated cost critic and improves held-out action-scale AUROC from 0.387 to 0.530. These results support the hypothesis that learned deployment time risk scoring can add useful information beyond a safe RL cost critic.

1 Introduction

Safe reinforcement learning (RL) aims to train agents that maximize task reward while respecting safety constraints. A common formulation is the constrained Markov decision process (CMDP) where a policy optimizes expected return subject to a bound on expected cost. Algorithms such as constrained policy optimization and Lagrangian actor critic methods provide practical ways to train policies under these constraints [Achiam et al., 2017, Ray et al., 2019]. However satisfying a cost constraint during training does not guarantee safe behavior after deployment, especially under distribution shift.

This project studies a simple question at deployment time. Can a trained safe RL policy’s violation of a safety specification be predicted before the violation happens? And if so, can that risk score be used to gate the policy and switch to a conservative fallback? The main environment is the lightweight Safety-Gymnasium environment `SafetyPointGoal1-v0` where a point robot must navigate to goals while avoiding hazards [Ji et al., 2023]. The policy is trained with PPO-Lagrangian so it already learns a cost value function $V_c(s)$. This makes $V_c(s)$ a natural and strong baseline. If the learned risk head can’t outperform the cost critic then it is not adding value at deployment.

The main contribution is a compact end to end pipeline for this question. It trains a PPO-Lagrangian policy from scratch, collects shifted rollouts as direct sampling falsification data, trains a risk head on runtime features, calibrates the cost critic baseline and evaluates both calibration and selective deployment. The implementation avoids big RL libraries so that the method and baselines can still be transparent.

2 Related Work

CMDPs give a standard formalism for reward maximization under cost constraints [Altman, 1999]. PPO-Lagrangian methods adapt policy-gradient optimization by subtracting a learned multiplier times the cost advantage from the reward advantage. This produces a simple constrained variant of PPO [Schulman et al., 2017]. Constrained policy optimization provides a more explicit trust region alternative [Achiam et al., 2017]. Safety Gym and Safety-Gymnasium provide benchmark environments for studying these algorithms [Ray et al., 2019, Ji et al., 2023].

The risk modeling setup is also related to validation and falsification. In the terminology of Kochenderfer et al. [2021] (Stanford professor and research advisor of mine) distribution shifts can be viewed as disturbance distributions, which are random variables that perturb sensors, agents or environment dynamics. Sampling rollouts under these disturbances is a direct sampling falsification procedure. Unlike optimization based falsification, this project does not search adversarially for failures but rather uses sampled shifted trajectories to learn and evaluate risk predictions at deployment.

Finally the cost critic baseline is calibrated with Platt scaling [Platt, 1999]. This turns a scalar score into a probability via logistic regression. Calibration matters here because the selective deployment thresholds should be comparable between the learned risk head and the cost critic.

3 Method

3.1 Base Safe Policy

The base policy is trained as a CMDP. The objective is to maximize expected episodic reward while keeping expected episodic cost below

$$C_{\text{limit}} = 25.$$

The actor is a two-layer MLP with tanh activations and a log standard deviation vector that is independent of the state. Actions are sampled from a Gaussian distribution and clipped to the bounds of the environment before stepping. Two separate critics estimate reward value $V(s)$ and cost value $V_c(s)$.

For each rollout, generalized advantage estimation is computed separately for rewards / costs. The Lagrange multiplier $\lambda \geq 0$ is updated after each rollout using the completed episodes' mean cost:

$$\lambda \leftarrow \text{clip}(\lambda + \eta_\lambda(\bar{C} - C_{\text{limit}}), 0, \lambda_{\text{max}}).$$

The policy advantage used by PPO is

$$A(s, a) = \frac{A_r(s, a) - \lambda A_c(s, a)}{1 + \lambda},$$

followed by the standard PPO clipped surrogate objective [Schulman et al., 2017].

3.2 Risk Labels

The deployment-time safety label is defined over a short look ahead window rather than a full episode. For each step t , let

$$C_t = \sum_{k=t}^{\min(t+H, T)} c_k,$$

where $H = 50$ and T is the end of the episode. A step is labeled risky if $C_t > B$. The robustness margin is $\rho_t = B - C_t$. The pipeline can use a fixed $B = 1.0$ but by default it selects an effective budget from in-distribution window costs to avoid severe label imbalance.

3.3 Risk Head and Cost-Critic Baseline

The risk head predicts $P(y_t = 1)$ from features available at deployment:

$$x_t = [o_t, \mu_\pi(o_t), \log \sigma_\pi, V(o_t), V_c(o_t), \text{recent-cost}_t].$$

It is a small MLP with two hidden layers of width 64, ReLU activations, feature normalization and a sigmoid output. Since positives can be rare, training uses weighted binary cross-entropy loss.

The baseline uses $V_c(s)$ directly as a risk score. To make thresholds comparable, a 1D logistic regression is fit on $V_c(s)$ using the same training split. This Platt-scaled baseline outputs a calibrated probability and is more meaningful than raw cost-value thresholding.

3.4 Disturbance Distributions and Fallback

The experiments use three disturbance families. Observation noise adds Gaussian noise to observations and represents a sensor disturbance. Action noise adds Gaussian noise before stepping and represents an agent disturbance. Action scaling multiplies action before stepping and represents an environment or dynamics disturbance. The risk head trains only on nominal, observation noise and action noise episodes. Action scaling is held out as OOD disturbance family.

Selective deployment uses a probability threshold τ . If risk exceeds τ the agent takes a conservative fallback action. Otherwise it takes the policy action. The fallback is zero action (does not move). This tests whether risk prediction alone can make a minimal intervention useful without training a second recovery policy.

4 Experimental Setup

All experiments use `SafetyPointGoal1-v0`. The final evaluation trains PPO-Lagrangian for 20,000 steps and collects four episodes per shift. This setting keeps the study focused on the deployment time risk-scoring question instead of obtaining a state-of-the-art base policy.

The in-distribution dataset has nominal rollouts, observation noise with standard deviations 0.02 and 0.05 and action noise with standard deviations 0.05 and 0.10. Episodes are split by episode ID so no trajectory leaks across train and test. The OOD dataset is the held out action scaling shift with factor 1.4. Evaluation uses AUROC, AUPRC, Brier score and expected calibration error with ten bins. For selective deployment the comparison is ungated PPO-Lagrangian, risk-head gating, cost-critic gating and always fallback.

5 Results

5.1 Calibration and Predictiveness

Table 1 summarizes aggregate calibration and predictiveness. The risk head improves the in-distribution AUROC from 0.907 to 0.956 and AUPRC from 0.466 to 0.751. This is evidence that the learned head is not just copying the PPO-Lagrangian cost critic. It is using the extra runtime features, especially the policy distribution and recent cost, and it allows to better separate risky from non-risky windows. On the held-out action scale shift, the risk head improves AUROC from 0.387 to 0.530 and AUPRC from 0.096 to 0.260. The OOD result is much weaker than the in-distribution result, which is expected since action scaling is a disturbance family that is absent from risk head training.

Figure ?? shows the same. In the ROC panel the in-distribution curve for the risk head stays above the cost critic for most false positive rates. This means the gain is not only due to one threshold choice. The OOD panel is flatter and closer to chance. However the risk head still has a visible advantage over the cost critic. The calibration panel is more mixed. The risk head has lower ECE than the cost critic on both the in-distribution split (0.101 vs 0.360) and the held-out action-scale split (0.126 vs 0.389) but the reliability curves are not diagonal perfectly. In particular many examples have very low predicted risk while the empirical positive rate is not zero. So the risk head is better calibrated than the baseline, but it is still underconfident in some of the low risk bins.

Table 1: Calibration and predictiveness. The effective short-window risk budget is $B = 0.5$.

Split	Model	Pos. rate	AUROC	AUPRC	Brier	ECE
ID test	Risk head	0.122	0.956	0.751	0.099	0.101
ID test	Cost critic	0.122	0.907	0.466	0.225	0.360
OOD	Risk head	0.128	0.530	0.260	0.126	0.126
OOD	Cost critic	0.128	0.387	0.096	0.268	0.389

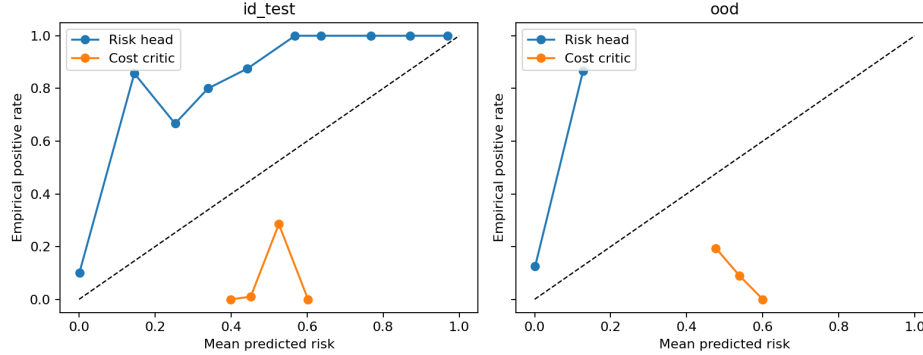


Figure 1: Calibration curves for the learned risk head and Platt-scaled cost critic.

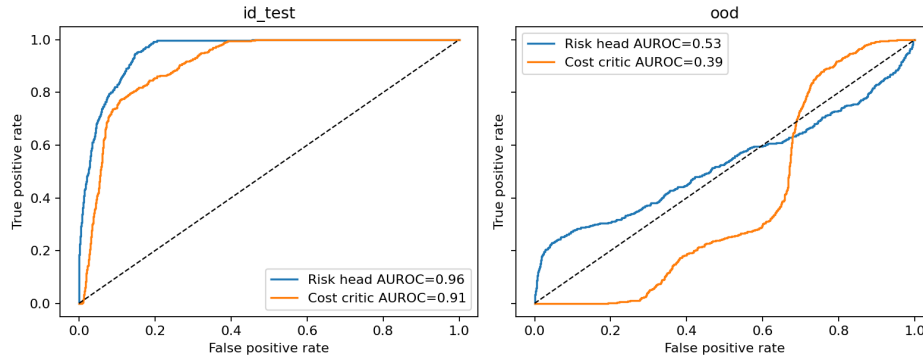


Figure 2: ROC curves for the learned risk head and Platt-scaled cost critic.

5.2 Per-Family Breakdown

Table 2 breaks results out by disturbance family. This is more informative than only reporting aggregate rows because the risk head’s advantage is not uniform. The largest gain appears on action noise (where AUROC improves from 0.580 to 0.974). This makes sense because action noise changes the realized behavior while leaving the policy’s intended action distribution available as a useful runtime signal. Observation noise also shows a large gain (from 0.745 to 1.000) although its positive rate is only 0.030 so the perfect score should be interpreted carefully. The held-out action-scale family still shows a positive AUROC gain 0.143, but absolute performance is close to chance. This tells us that the learned head generalizes somewhat to a new disturbance family but dynamics-like shifts is still the hardest part of the project.

5.3 Selective Deployment

Figure 3 shows the reward-cost tradeoff when risk probabilities gate the base policy. The plot annotates the threshold values swept by each gating method. On the observation-noise shift, both gating methods find settings with zero mean cost, but the cost-critic gate uses the fallback much more often

Table 2: Per-disturbance-family predictiveness. AUROC gain is risk head minus Platt-scaled cost critic.

Family	Pos. rate	Head AUROC	Critic AUROC	Gain	Head AUPRC	Critic AUPRC
Action noise	0.166	0.974	0.580	0.394	0.912	0.183
Action scale	0.128	0.530	0.387	0.143	0.260	0.096
Nominal	0.051	1.000	0.473	0.527	1.000	0.044
Observation noise	0.030	1.000	0.745	0.255	1.000	0.072

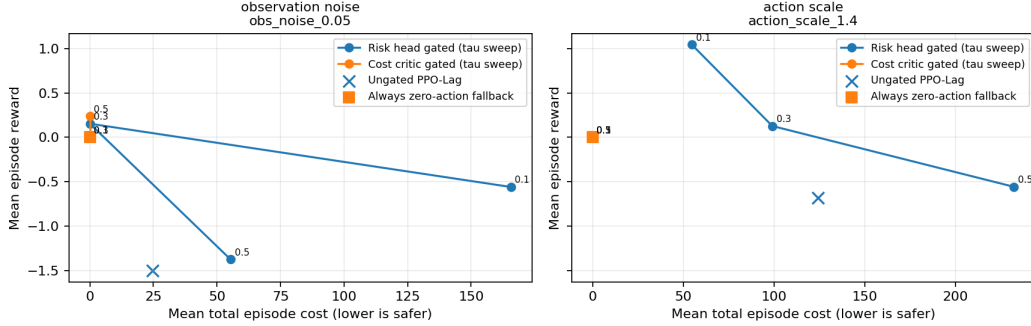


Figure 3: Selective deployment reward-cost tradeoff. Lower cost is safer and higher reward is better. Curves sweep the risk threshold τ . Marked points show ungated PPO-Lagrangian and always-fallback.

(0.83 fallback fraction at its selected threshold) than the risk head (0.40). This is useful because it shows the difference between being safe and being overly conservative. On the action-scale shift, the risk head gate keeps more reward at one threshold but pays for it with much higher mean cost and a nonzero violation rate. The cost critic gate is more conservative there and often collapses toward the always-fallback behavior. The main lesson from this panel is that risk prediction quality and fallback design are both needed for good selective deployment.

6 Discussion

The main positive result is that the risk head can outperform a calibrated cost critic even though $V_c(s)$ is already a strong baseline. This matters because $V_c(s)$ is not a weak strawman. It is the cost value function learned by the PPO-Lagrangian agent itself and it should contain information about future safety costs. The fact that the risk head improves over it means that risk at deployment time is not fully captured by the scalar cost critic. The added features (especially recent cost and policy distribution information) appear to help see /identify short-horizon failure windows.

The calibration result is also important. The cost critic baseline has noticeably worse Brier score and ECE. This means only using Platt scaling does not fully turn $V_c(s)$ into a reliable deployment probability. The risk head is not perfectly calibrated either but it is closer to the empirical risk curve. This is important for selective deployment because the threshold τ only has a clear meaning if predicted probabilities are at least roughly calibrated. A model with high AUROC but bad calibration could still be hard to use as a gate.

The per-family results show the main limitation of the approach. The risk head does very well on nominal, observation-noise and action-noise data. However the held-out action-scale shift is much harder. This is the most interesting failure mode. Observation noise and action noise mostly perturb signals around the learned policy while action scaling changes how actions map into movement. That makes the transition dynamics different in a way that the runtime features do not fully show us. In other words the risk head learns useful correlates of risk but does not learn a general dynamics model.

The selective-deployment results should be read similarly. The zero action fallback is intentionally simple. It can reduce cost by stopping the agent but it can also lose reward or fail after the agent is already in a hazard. The observation-noise panel is encouraging because the risk head reaches low cost with less fallback use than the cost critic. The action-scale panel is more mixed. There the risk head can preserve reward but sometimes lets too much cost through, while the cost critic tends to be safer but more conservative. So the project supports the usefulness of risk scoring, but also shows that a better fallback policy would matter a lot.

7 Conclusion

This work built and verified a complete pipeline for deployment-time risk scoring in safe RL. The pipeline trains a PPO-Lagrangian policy, collects direct-sampling falsification data under disturbance distributions, trains a risk head, calibrates the cost critic baseline and evaluates calibration plus selective deployment. The learned risk head improves over the Platt-scaled cost critic both in-distribution and on the held-out action-scale disturbance. The final result satisfies the success criterion defined of at least 0.10 AUROC improvement over the calibrated cost critic on the held-out disturbance family.

The bigger takeaway is that a safe RL cost critic is a useful baseline but should not be treated as the whole answer to safety monitoring at deployment. A small learned head using runtime features can produce better short-horizon risk scores and better calibrated probabilities. The strongest evidence appears under the disturbance families represented in the risk dataset, while the held-out action-scale setting shows that OOD generalization is still limited. Overall the project gives a practical recipe for using direct-sampling falsification data to build a lightweight deployment monitor and it also makes clear that risk scoring and fallback design need to be evaluated together.

8 AI Tools Disclosure

Per the guidelines I used AI tools to help with boilerplate code, debugging and helping organize the experiment. The core project decisions, safe RL formulation, interpreting results and understanding of the PPO-Lagrangian algorithm did not use AI.

References

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A Additional Per-Shift Results

Table 3: Per-shift predictiveness.

Shift	Pos. rate	Head AUROC	Critic AUROC	Head AUPRC	Critic AUPRC
action_noise_0.05	0.209	1.000	0.649	0.998	0.345
action_noise_0.10	0.122	0.934	0.585	0.734	0.130
action_scale_1.4	0.128	0.530	0.387	0.260	0.096
nominal	0.051	1.000	0.473	1.000	0.044
obs_noise_0.02	0.034	1.000	0.696	1.000	0.056
obs_noise_0.05	0.026	1.000	0.825	1.000	0.176